BAHRAIN POLYTECHNIC بوليتكنك البحرين

IT9204: Emerging Technologies in AI

Individual Project – Topic Approval Form

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Note: please keep your form answers short and concise. The purpose of this form is to ensure your selected topic is feasible for the nature of the project, and that you have a plan on how it can be implemented. Actual design and implementation decisions and details are expected within the final submission only.

Project Scenario

Select one of the provided project scenarios (e.g., Student Success Assistant, HR Attrition Assistant) or propose a custom scenario of equivalent complexity.

Custom scenario - Financial Fraud Detection and Advisory Assistant. Equivalent in complexity to Scenario 5: Financial Risk and Loan Advisory Assistant.

Problem Description

Clearly describe the problem your system will solve and the type of user it is designed for (e.g., academic advisor, HR manager).

Banks and financial institutions face significant losses from fraudulent transactions. This system assists fraud analysts and bank risk officers by combining predictive fraud detection with policy-grounded decision support. Given a transaction or customer query, the system predicts fraud risk, retrieves relevant compliance and response policies, and recommends a course of action.

RAG Data Source (Documents)

Describe the documents you will use (e.g., policies, manuals, guidelines). Include source (URL), type, and approximate size/number of documents.

- Central Bank of Bahrain fraud reporting guidelines - https://www.cbb.gov.bh (PDF, ~3-5 documents)
- PCI-DSS compliance standards - https://www.pcisecuritystandards.org (publicly available PDF, ~50 pages)
- Bank fraud response and escalation procedures (synthetic document, ~5-10 pages)
- Anti-money laundering and KYC policy guidelines (publicly available, ~10-15 pages)

Total: approximately 4-6 documents, mixture of regulatory and procedural content.

Structured Data / Tools

Define the prediction problem (e.g., risk prediction, classification). Specify target variable and input features.

Dataset: Kaggle Credit Card Fraud Detection dataset URL: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

Key features: transaction amount, time, 28 anonymized PCA features (V1-V28), and Class label (0 = legitimate, 1 = fraud). Contains 284,807 transactions with a 0.17% fraud rate - representative of real-world class imbalance.

Tools the agent will use:

- `predict_fraud_risk(transaction_features)` - calls the deployed ML endpoint
- `retrieve_policy(query)` - RAG tool searching compliance and fraud policy documents
- `calculate_risk_score(amount, frequency, location_change, time_of_day)` - custom function computing a composite risk indicator

Machine Learning Task

Define the prediction problem (e.g., risk prediction, classification). Specify target variable and input features.

Binary classification: predict whether a transaction is fraudulent (Class = 1) or legitimate (Class = 0).

Target variable: Class (binary) Input features: transaction Amount, Time, and V1-V28 (PCA-transformed features)

Machine Learning Plan

Briefly describe how you will use Machine Learning (e.g., classification/regression, dataset preparation, expected output).

Train a classification model (Random Forest or XGBoost) on the Kaggle dataset using Azure Machine Learning AutoML. Handle class imbalance using SMOTE oversampling or class weights. Evaluate using precision, recall, F1-score, and AUC-ROC. Deploy the trained model as an Azure ML real-time endpoint. The agent calls this endpoint as a tool, passing transaction features and receiving a fraud probability score and predicted class.

Agent Functionality

Explain what your agent will do and what tools it will have access to (e.g., retrieve documents, call ML model, query data).

The agent acts as a fraud advisory assistant for bank risk officers. It has access to three tools:

1. `predict_fraud_risk` – calls the Azure ML endpoint to get a fraud probability for a given transaction
2. `retrieve_policy` – searches the RAG knowledge base for relevant fraud response procedures, compliance rules, and escalation guidelines
3. `calculate_risk_score` – a deterministic function that computes a composite risk indicator from transaction metadata

The agent decides which tools to invoke based on the user's query, combines the outputs, and provides a grounded, explainable recommendation.

Decision Logic (Critical)

Explain how your system will decide when to use RAG, tools, or the ML model (high-level explanation only).

- Query about a specific transaction – call `predict_fraud_risk`. If probability exceeds 0.7, also call `retrieve_policy` to fetch escalation procedures. Combine both into the final response.
- Query about what action to take – call `retrieve_policy` only. ML prediction not needed.
- Query about a pattern of transactions – call `calculate_risk_score` first. If composite score is high, follow with `predict_fraud_risk` and `retrieve_policy`.
- Ambiguous queries – agent reasons through context and selects the most appropriate tool(s).

The agent never fabricates policy rules. All recommendations are grounded in retrieved documents.

Example User Queries

Provide 2–3 example user queries that demonstrate how your system will be used.

1. "A transaction of $4,500 was made from a new device at 2:30 AM. Is this likely fraud?"
2. "What is the bank's policy when a high-risk transaction is detected on a corporate account?"
3. "Customer Ahmed made 14 transactions in 20 minutes totalling $6,200. What does the risk assessment say and what action should be taken?"